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Investigation on chemical structures of premixed toluene flames at low pressure

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Abstract

Chemical structures of premixed toluene/O_2/Ar flames with five equivalence ratios (0.75, 1.00, 1.25, 1.50, and 1.75) were studied at low pressure (4.0 kPa). Synchrotron vacuum ultraviolet photoionization mass spectrometry was used for the isomeric detection of flame species and the measurements of their mole fractions. The global trends of the flame temperature and mole fractions of detected species with varying equivalence ratio were observed and discussed, drawing complete pictures for the chemical structures of premixed toluene flames. Based on the experimental results, a kinetic model was developed from previous models and updated with many recently studied reactions related to toluene decomposition and polycyclic aromatic hydrocarbons (PAHs) formation. The model was validated by simulating the measured mole fractions of flame species, showing good agreement in reproducing the maximum mole fractions of most observed species and their global trends with varying equivalence ratio. The rate of production analysis reveals the major formation and consumption channels of some key intermediates involved in toluene decomposition in the $\phi = 0.75$ and 1.75 flames, and the major formation channels of some large aromatics and typical PAHs in the $\phi = 1.75$ flame, indicating the importance of benzyl in toluene flames.
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Keywords: Premixed toluene flame; Synchrotron photoionization; Kinetic model; Toluene decomposition; PAHs formation

1. Introduction

Understanding the combustion chemistry of monocyclic aromatic hydrocarbons (MAHs) is of special interest, because they are important components in practical fuels such as gasoline, diesel oils, and jet fuels, and also because of the convenience in studying the formation mechanism of polycyclic aromatic hydrocarbons (PAHs) and soot [1]. As the simplest alkylated aromatic hydrocarbon, toluene is typically used in gasoline reference fuels to simulate the combustion behaviors of aromatic components [2–3], and serves as a prototype fuel in the development of kinetic models for

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large aromatic surrogates of diesel and jet fuels. On the other hand, toluene is a common intermediate in flames of large aromatics; and its flame is the best system to study the chemistry of benzyl which plays a significant role in fuel decomposition and PAHs formation processes in flames of alkylated aromatics.

Because of the desirable properties such as the low boiling point and weak toxicity compared with other aromatics, toluene is one of the most experimentally investigated aromatic fuels as summarized in [4–5], especially in recent three years [2–11]. The experimental results were used to validate the kinetic models for toluene combustion [4,7,9,11–12] in recent years. Meanwhile, the great concerns on experimental and modeling studies of toluene combustion gathered a lot of efforts on the theoretical studies of reactions related to toluene decomposition [10,12–18]. However, most previous experimental studies were focused on measurements of global combustion properties, and thereby only provided limited chemical information; the other ones reporting measurements of chemical structures were mostly performed at only one condition, which is not convenient for simultaneous validation of both oxidation and pyrolysis submechanisms.

In this work, an experimental investigation was performed on the chemical structures of premixed toluene/O_2/Ar flames over a broad range of equivalence ratios ($\phi = 0.75$–1.75) at low pressure, with the application of synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) in the identification of flame species and quantification of their mole fractions. The variation of equivalence ratio exhibits the pictures of toluene chemistry from oxidation dominated to pyrolysis dominated circumstances. Based on the experimental work, a kinetic model was developed from previous models and updated with many recently studied reactions related to toluene decomposition and PAHs formation.

2. Experimental specifications

Experimental work was performed at National Synchrotron Radiation Laboratory at Hefei, China. Detailed description of the experimental instruments was published elsewhere [5,19]. The flame conditions are listed in Table 1, showing that five flames with increasing equivalence ratios and carbon-to-oxygen (C/O) ratios were studied at 4.0 kPa. For a better comparison of chemical issue, the pressure, inlet percentage of Ar, and cold gas velocity of inlet mixture were kept identical in all flames. All measurements were performed at the axial direction of the burner. Methodologies of intermediate identification and mole fraction evaluation were reported previously [5,20]. The uncertainties of evaluated mole fractions are within ±10% for major species, ±25% for intermediates with accurately known photoionization cross sections (PICS), and a factor of 2 for those with estimated PICS [5]. Recently, the PICS of many aromatic hydrocarbons were measured using SVUV-PIMS [21], and were used in this work. The flame temperature was measured using a Pt-6%Rh/Pt-30%Rh thermocouple with a diameter of 0.100 mm. The thermocouple was coated with Y_2O_3–BeO anti-catalytic ceramic to inhibit the catalytic effects [22–23]. The temperature was corrected for the radiation heat loss [24] and cooling effects of sampling nozzle [25]. The uncertainty of the maximum flame temperature was estimated to be ±100 K.

Table 1
Flame conditions.

ϕ	C/O	P/kPa	X_{Fuel}	X_{O2}	X_{Ar}	V/cm s^{-1}
0.75	0.29	4.0	3.85	46.15	50.00	35.00
1.00	0.39	4.0	5.00	45.00	50.00	35.00
1.25	0.49	4.0	6.10	43.90	50.00	35.00
1.50	0.58	4.0	7.14	42.86	50.00	35.00
1.75	0.68	4.0	8.14	41.86	50.00	35.00

Note: X_i is the inlet percentage of species i; V is the flow velocity of inlet mixture at 300 K.

3. Kinetic modeling

Based on the measured temperature profiles, the simulation was performed by using the CHEMKIN II and PREMIX code [26–28]. The model developed in this work includes 176 species and 804 reactions. Table S1 in Supplemental data lists a series of pyrolysis and oxidation reactions related to toluene decomposition and PAHs formation in this model, and the references of their rate coefficients as well. Many of them were recently studied with updated rate coefficients, particularly those participating in toluene decomposition. In brief, this model was developed from our recently reported model for low-pressure toluene pyrolysis [29], which consists of pyrolysis reactions mainly drawn from four previous models, that is, the USC Mech II model for the C_1–C_6 reactions [30] and the models reported by Wang and Frenklach [31], Appel et al. [32], and Richter et al. [33] for aromatic reactions. Therefore detailed descriptions of pyrolysis reactions in this model can be found in Ref. [29].

On the other hand, the oxidation reactions in this model were mainly drawn from four above-mentioned models [30–33], and the models for toluene combustion reported by Bounaceur et al. [34], Andrae et al. [35], Blanquart et al. [36] etc. Furthermore, recently studied oxidation reactions of toluene and its primary decomposition products were updated in this model, such as the reactions of OH attack on toluene (R6–R8 in

Table S1) studied by Seta et al. in 2006 [37]. They measured the rate constants of these reactions directly by a shock tube/pulsed laser-induced fluorescence imaging method at high temperatures. The reaction mechanism, thermodynamics data, and gas transport data can be found in Supplemental data.

4. Results and discussion

4.1. Flame temperature and major flame species

Figure 1(a) shows the axial temperature profiles of all five flames. Each profile increases from the low temperature of inlet mixture near the burner surface to the maximum temperature (T_{max}), and then drops as the distance increases because of the heat loss. The accuracy of temperature measurements can be examined by the following approaches. First, the T_{max} increases from $\phi = 0.75$ to 1.25 and decreases from 1.25 to 1.75, which is consistent with the trend of adiabatic flame temperature of toluene/O_2/50%-Ar mixture as seen from Fig. 2. Secondly, the distance of the T_{max} position from the burner surface, abbreviated as DFB_{max} in this work, decreases from $\phi = 0.75$ to 1.00 and increases from $\phi = 1.00$ to 1.75, which is opposite to the trend of laminar flame speed reported by Davis and Law [38] as shown in Fig. 3. Recognizing that the location of reaction zone is related to the position of the high temperature region, the motion of the reaction zone with varying equivalence ratio can be revealed. Since the cold gas velocity is identical for all flames, the flame with higher laminar flame speed needs more heat loss to the burner plate and consequently smaller DFB_{max} to balance with the cold gas velocity according to the study of Botha and Spalding [39].

Figure 1 also shows the measured mole fraction profiles of major flame species in all flames, including reactants (toluene and O_2), inert gas (Ar), and major products (H_2, H_2O, CO, and CO_2). In each flame, the mole fraction of Ar drops downstream first and keeps almost invariable afterwards, as a result of the positive mole expansion effects of toluene flame; and the fall-off trend becomes more salient with increasing equivalence ratio, as expected. The predicted mole fraction profiles of Ar exhibit good agreement with the experimental results, indicating that this model accurately reproduced the mole expansion effects over the broad range of equivalence ratio.

In each flame, the mole fractions of toluene and O_2 keep falling in the preheat zone and the reaction zone. Toluene is depleted in all flames, while O_2 is completely consumed only in $\phi = 1.75$ and 1.50 flames and partly consumed in other flames because of the thermochemical equilibrium. The residual O_2 is observed to increase with decreasing equivalence ratio. We see that this

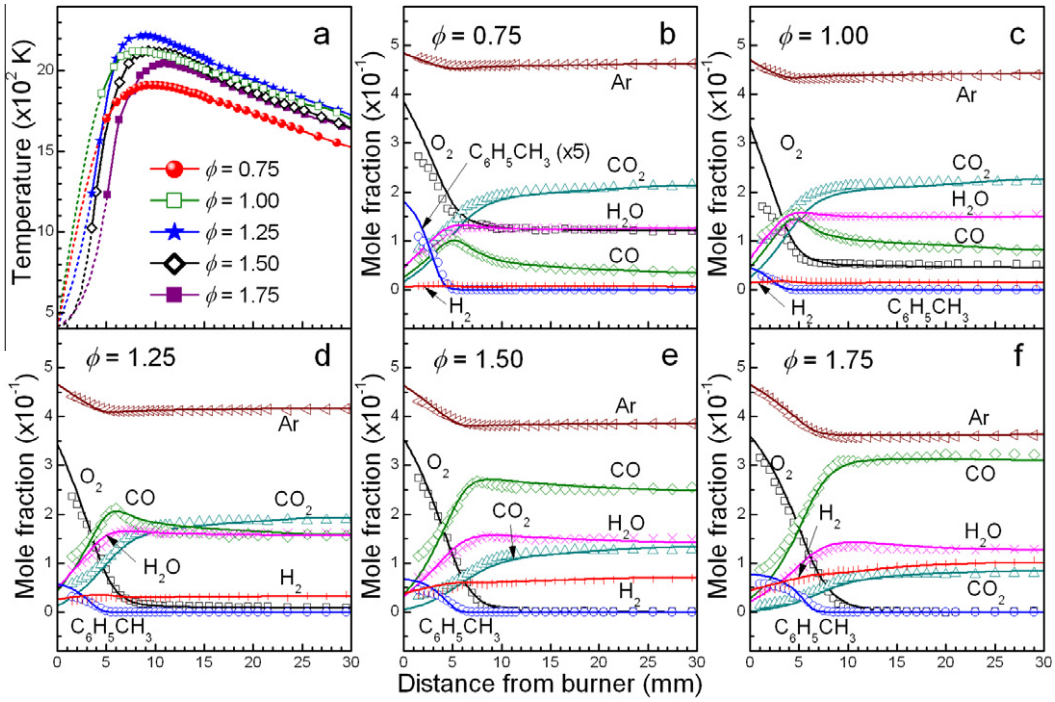

Fig. 1. (a) Temperature profiles of five toluene flames investigated; (b–e) Measured (symbols) and predicted (solid lines) mole fraction profiles of major flame species in $\phi = 0.75$, 1.00, 1.25, 1.50, and 1.75 flames, respectively.

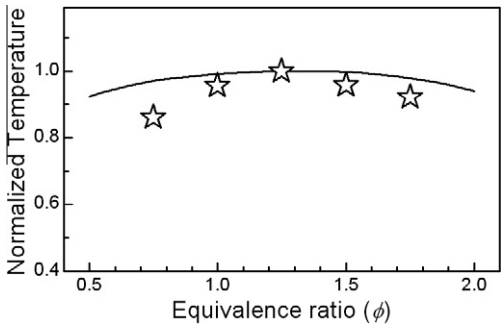

Fig. 2. Comparison of normalized maximum flame temperature (T_{max}/T_{max} ($\phi = 1.25$), stars) with normalized adiabatic flame temperature (T_{ad}/T_{admax}, solid line) of toluene/O_2/50%-Ar mixture at 30 Torr.

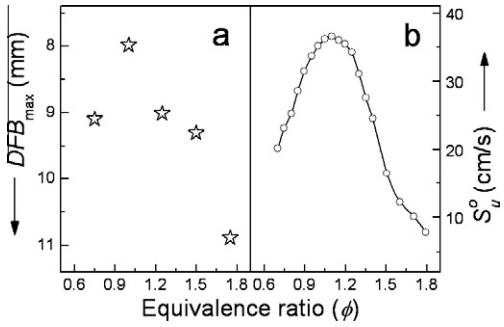

Fig. 3. (a) Plot of DFB_{max} (stars) versus the equivalence ratio. (b) Laminar flame speeds of toluene/air mixture at 1 atm (solid line with circles) reported by Davis and Law [38] which reveal the peak shape of laminar flame speed as a function of equivalence ratio.

model quantitatively predicts the consumption of reactants, particularly their depletion positions and the equilibrium mole fractions of O_2.

H_2O is the dominant hydrogenous product in all flames except for the richest one, while H_2 has rising equilibrium mole fractions with increasing equivalence ratio and has comparable equilibrium mole fraction with H_2O in the richest flame. CO has similar global trend with H_2, and gradually replace CO_2 to be the major carbonous products as equivalence ratio increases. It is found that the stoichiometric flame has the highest equilibrium mole fractions of H_2O and CO_2. However if normalized by inlet carbon or hydrogen flux, the mole fractions of H_2O and CO_2 vary oppositely to those of H_2 and CO with increasing equivalence ratio, as shown in Fig. 4. The comparison in Fig. 1 indicates that this model reproduces the mole fraction profiles of these major products in satisfactory agreement.

4.2. Toluene decomposition and small intermediates

Dozens of intermediates, including a series of isomers and radicals, were identified in this work.

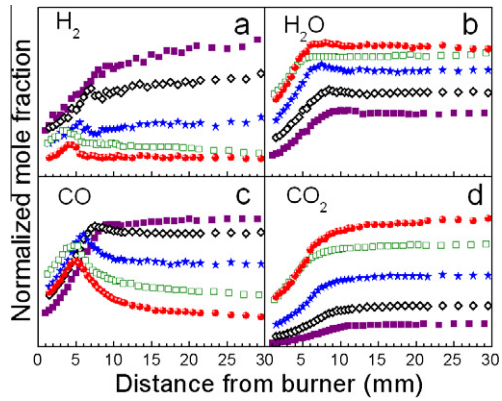

Fig. 4. Mole fraction profiles of (a) hydrogen and (b) water normalized by inlet hydrogen flux, and (c) carbon monoxide and (d) carbon dioxide normalized by inlet carbon flux. Symbols: $\phi = 0.75$ – solid circles, 1.00 – squares, 1.25 – solid stars, 1.50 – diamonds, 1.75 – solid squares.

The illustrations of intermediate identification in toluene flames can be found in our previous studies [5,20]. Most intermediates with molecular weight smaller than 92 Da were detected to be identical in all five flames, while the number and concentrations of aromatic species with molecular weight greater than 92 Da were found to increase with increasing equivalence ratio. For the purpose to compare the combustion chemistry in lean and rich toluene flames, the formation and consumption channels of some intermediates are discussed below for $\phi = 0.75$ and 1.75 flames.

The primary decomposition of toluene can produce many intermediates, such as benzyl, benzene, phenyl, methylphenyl, and methylphenoxy. The rate of production (ROP) analysis indicates that the H-abstraction channels of toluene by H, O, and OH attack dominates the primary decomposition of toluene in both $\phi = 0.75$ and 1.75 flames. In the $\phi = 0.75$ flame, O and OH attack channels consumes more than 70% of toluene, while H attack channels only have minor contributions. By contrast, the H attack channels, particularly the one to produce benzyl and H_2, control the consumption of toluene in the $\phi = 1.75$ flame with a percentage of more than 70%.

Figure 5 illustrates the mole fraction profiles of six intermediates with molecular weight smaller than 92 Da, which play significant roles in toluene decomposition. The uncertainties of measured mole fractions are ±25% for benzene, vinylacetylene, and acetylene, and a factor of 2 for benzyl, cyclopentadienyl, and propargyl. Benzyl has high concentrations in toluene flames, for example, its maximum mole fractions (X_{max}) in the $\phi = 0.75$ and 1.75 flames are measured to be 1.8×10^{-4} and 1.3×10^{-3}, respectively. It is the most important product from the primary decomposition of

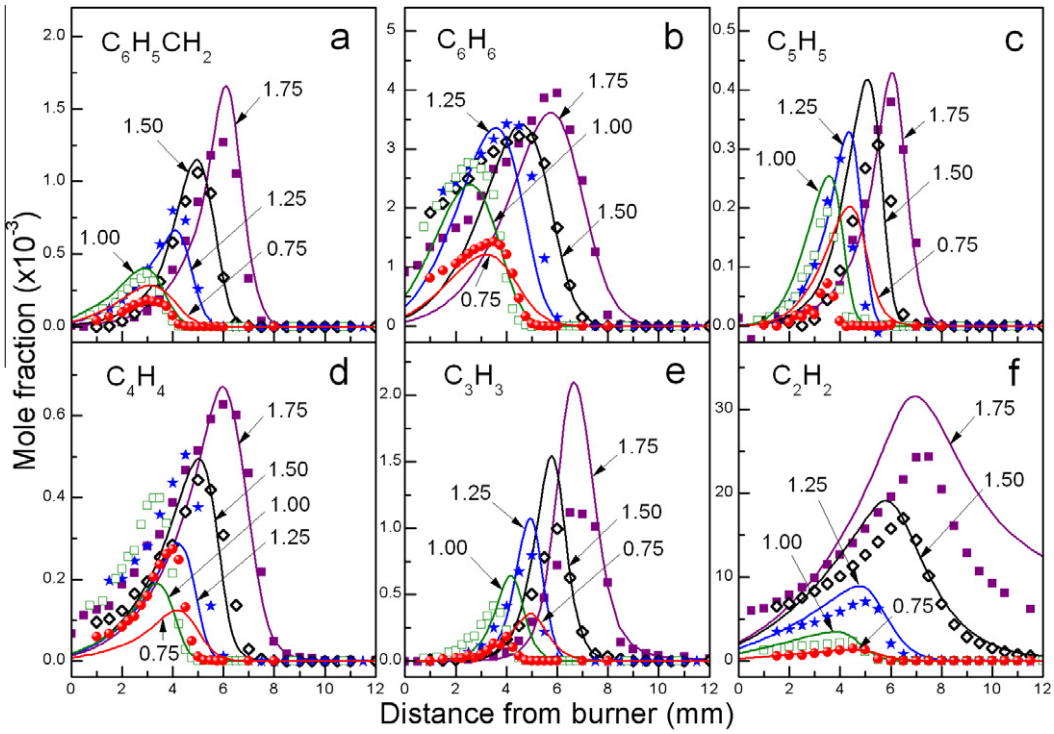

Fig. 5. Measured and predicted mole fraction profiles of (a) benzyl, (b) benzene, (c) cyclopentadienyl, (d) vinylacetylene, (e) propargyl, and (f) acetylene in all five toluene flames. Experimental results: $\phi = 0.75$ – solid circles, 1.00 – squares, 1.25 – solid stars, 1.50 – diamonds, 1.75 – solid squares. Simulation results are marked with respective equivalence ratios.

toluene, affording more than 60% and 70% of toluene consumption in the $\phi = 0.75$ and 1.75 flames, respectively. This is reasonable because of the weakest bond dissociation energy of the methyl C–H bond among the C–H and C–C bonds in toluene molecule, and the lowest enthalpy of formation of benzyl among all C_7H_7 radicals as well [40]. Figure 5(a) shows that the model reproduced the measured X_{max}s of benzyl in all flames and the dropping trend of its mole fraction with decreasing equivalence ratio. The ROP analysis indicates that benzyl is mainly formed from the H-abstraction of toluene by OH attack in the $\phi = 0.75$ flame and by H attack in the $\phi = 1.75$ flame.

Benzene is the other primary decomposition product of toluene which can be experimentally detected in this work. As seen from Fig. 5(b), the predicted X_{max}s of benzene consistently fit with the measured data in all flames. The formation of benzene is dominated by the methyl replacement of toluene via H attack in both flames. This channel, however, only contributes around 10% for total toluene consumption in both flames.

The following four species in Fig. 5 are all produced in subsequent decomposition process with carbon atoms decreasing one by one. We see that the simulation results are in good agreement with the experimental results. In the $\phi = 0.75$ flame, the unimolecular decomposition of phenoxy produces about 50% of cyclopentadienyl. In the $\phi = 1.75$ flame, cyclopentadienyl is mainly formed from the H attack on fulvenallene which was detected to be the dominant C_7H_6 species in toluene flames [5,20] and suggested to be an important product from benzyl decomposition [9,17,41]. Further reaction sequences of cyclopentadienyl lead to the formation of vinylacetylene in both flames. Also in both flames, more than 60% of propargyl is generated from the unimolecular decomposition of cyclopentadienyl and the reaction between phenyl and H. Propargyl can produce many smaller C_1–C_3 molecules, particularly triplet propargylene (C_3H_2) which affords more than 70% of propargyl consumption in both flames. Acetylene is mainly produced from pyrolysis reactions such as the H attack on fulvenallene and the unimolecular decomposition of cyclopentadienyl in the $\phi = 1.75$ flame, while the O and OH attack on C_3H_2 has comparable contribution to acetylene formation in the $\phi = 0.75$ flame. The consumption of C_3H_2 and acetylene in both flames is controlled by oxidation reactions to produce oxygenated products including ketene (CH_2CO), ketenyl (HCCO), formyl (HCO) and

CO, which are involved in the formation of final oxidation products. In conclusion, oxidation reactions dominate toluene decomposition process in the $\phi = 0.75$ flame; by contrast, pyrolysis reactions, particularly the reaction sequence of toluene → benzyl → fulvenallene → cyclopentadienyl, have major contributions to toluene decomposition process in the $\phi = 1.75$ flame.

4.3. Large aromatics and PAHs

Figure 6 shows the mole fraction profiles of six aromatic intermediates with molecular weight greater than 92 Da for the discussion of aromatics growth and PAHs formation. The uncertainties of measured mole fractions are ±25% for phenylacetylene, ethylbenzene, and indene, and a factor of 2 for indenyl, naphthalene, and phenanthrene. Their mole fractions are all observed to drop with decreasing equivalence ratio because of the reduced toluene concentration in the inlet mixture, and more crucially due to the enhanced oxidation circumstance. The possible mechanisms relevant to aromatics growth and PAHs formation have been critically discussed in last two decades and were mainly proposed in two categories: one is the even carbon growth route or hydrogen-abstraction-carbon-addition (HACA) mechanism [42] including sequential addition channels of C_2 or C_4 species (acetylene, butadiyne, vinylacetylene, and their radicals) on PAH precursors; the other one is the odd carbon growth route driven by the addition of C_3 or C_5 resonantly stabilized radicals such as propargyl and cyclopentadienyl radicals on PAHs precursors [43–45], which is called as the resonantly stabilized radical addition mechanism in our previous work [5]. The experimental observations in this work indicate that acetylene, butadiyne, vinylacetylene, propargyl, and cyclopentadienyl all have high concentration levels in the rich flames. Therefore both mechanisms are included in this model.

For two C_8 MAHs, we see that the model predicts the X_{max}s of ethylbenzene satisfactorily, and reproduces the global trend of the X_{max}s for phenylacetylene. In both $\phi = 0.75$ and 1.75 flames, more than 80% of ethylbenzene is directly produced from the combination of benzyl and methyl, and it is mainly consumed by the H-loss reaction sequence to produce styrene. The consumption of styrene follows the H- and vinyl-abstraction by H attack in the $\phi = 1.75$ flame, while similar channels initiated by OH attack play a dominant role in the $\phi = 0.75$ flame. It is found that the H-abstraction of styrene leads to the formation of $C_6H_5CCH_2$ and C_6H_5CHCH radicals,

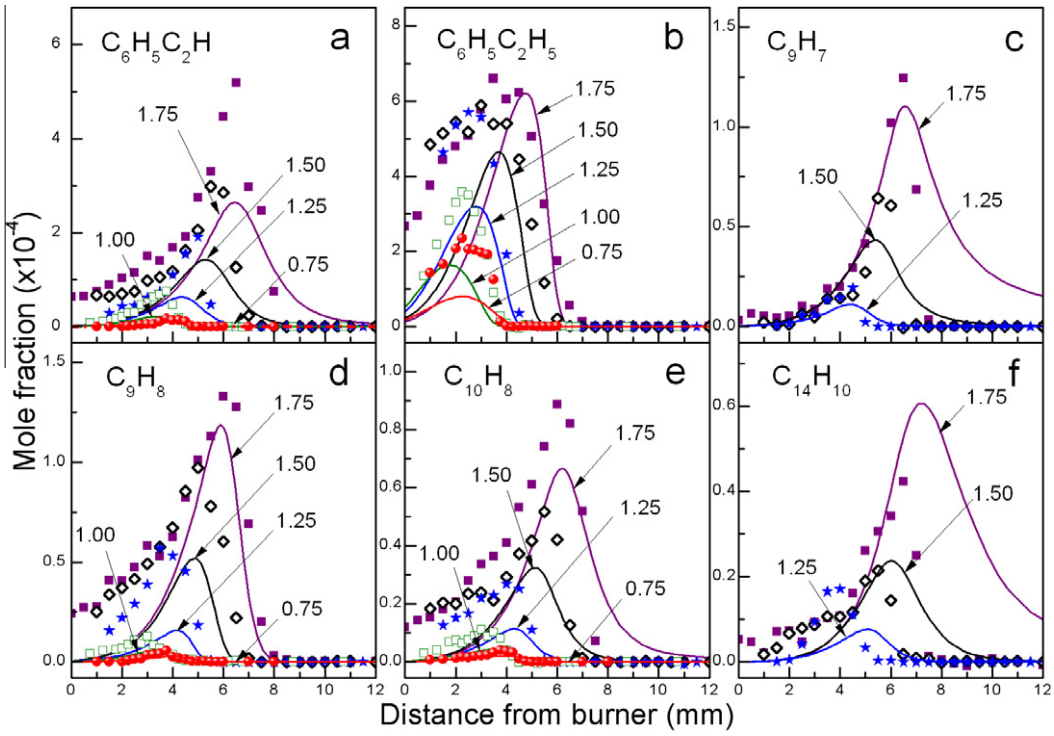

Fig. 6. Measured and predicted mole fraction profiles of (a) phenylacetylene, (b) ethylbenzene, (c) indenyl, (d) indene, (e) naphthalene, and (f) phenanthrene in all five toluene flames. Experimental results: $\phi = 0.75$ – solid circles, 1.00 – squares, 1.25 – solid stars, 1.50 – diamonds, 1.75 – solid squares. Simulation results are marked with respective equivalence ratios.

which are major precursors of phenylacetylene in both flames. Furthermore, the reaction of benzene and C_2H radical, a typical HACA channel, produces about 15% of phenylacetylene in the $\phi = 1.75$ flame.

The comparison in Fig. 6(c–f) shows that the model predicted the X_{max}s reasonably well for all four PAHs. Because PAHs only have high concentrations in the extremely rich flames, the discussion here is focused on the $\phi = 1.75$ flame. More than 70% of indene is formed via the reaction between benzyl and acetylene (a HACA channel) due to the high concentration of benzyl, while the propargyl addition on phenyl serves as a minor channel and produces about 20% of indene. Its consumption is governed by the H-loss reactions to produce indenyl. Indenyl has comparable X_{max}s with indene in three rich flames. However its concentrations in the lean and stoichiometric flames were beyond our detection limit due to its rapid oxidation.

The reactions relevant to naphthalene formation include the propargyl addition on benzyl, the reaction of phenyl and vinylacetylene, and the self-combination of cyclopentadienyl, etc. Also due to the high concentration of benzyl, the first reaction becomes the dominant formation channel of naphthalene in the $\phi = 1.75$ flame. Two isomeric tricyclic PAHs ($C_{14}H_{10}$), phenanthrene and anthracene, are detected in three rich flames, while no $C_{14}H_{10}$ species can be detected in the lean and stoichiometric flames. Recognizing that phenanthrene is the main $C_{14}H_{10}$ PAH, we present its mole fraction profiles in Fig. 6(f). Its formation channels in this model include the combination of indenyl and cyclopentadienyl, the reaction of phenylacetylene and phenyl, and the two-step H_2-loss channel of bibenzyl etc. Because bibenzyl is mainly formed from the recombination of benzyl, it has high concentrations not only in this work, but also in previous studies of toluene combustion [5,44]. Therefore the last channel contributes mostly to phenanthrene formation.

In conclusion, benzyl is major precursor of ethylbenzene, indene, naphthalene, and phenanthrene in the $\phi = 1.75$ flame, indicating that it plays a significant role in aromatics growth and PAHs formation process in rich toluene flames due to the high concentrations. It is aware that the possible mechanisms for PAHs formation are still under development among the combustion community. Therefore further experimental and theoretical investigations are essential to improve the performance in simulating PAHs formation. With consideration of the word limit, detailed analysis of the reaction flux at different equivalence ratios and further validation of this model with global combustion properties such as laminar flame speeds and ignition delay times will be reported in a subsequent modeling study.

5. Conclusion

Chemical structures of premixed toluene/O_2/Ar flames at low pressure (4.0 kPa) were studied over a broad range of equivalence ratio. Flame species were identified using SVUV-PIMS, and the mole fractions were evaluated. The global trends of the flame temperature and mole fractions of detected species were observed with varying equivalence ratios, drawing a complete picture for the chemical structures of premixed toluene flames. A kinetic model was developed from previous models and updated with many newly studied reactions related to toluene decomposition and PAHs formation. This model was validated based on the measured mole fractions of flame species. Good performance was observed in reproducing the mole fractions of most observed flame species and capturing their global trends with varying equivalence ratios. The ROP analysis reveals the major formation and consumption channels of some key intermediates involved in toluene decomposition and PAHs formation processes, indicating that benzyl plays a significant role in the fuel decomposition and PAHs formation processes in toluene flames.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.033.

References

[1] C.S. McEnally, L.D. Pfefferle, B. Atakan, K. Kohse-Hoinghaus, *Prog. Energy Combust. Sci.* 32 (2006) 247–294.
[2] L.R. Cancino, M. Fikri, A.A.M. Oliveira, C. Schulz, *Proc. Combust. Inst.* 32 (2009) 501–508.
[3] T. Bieleveld, A. Frassoldati, A. Cuoci, et al., *Proc. Combust. Inst.* 32 (2009) 493–500.
[4] A. El Bakali, L. Dupont, B. Lefort, N. Lamoureux, J.F. Pauwels, M. Montero, *J. Phys. Chem. A* 111 (2007) 3907–3921.
[5] Y.Y. Li, L.D. Zhang, Z.Y. Tian, et al., *Energy Fuel* 23 (2009) 1473–1485.
[6] M.A. Oehlschlaeger, D.F. Davidson, R.K. Hanson, *Proc. Combust. Inst.* 31 (2007) 211–219.
[7] G. Mittal, C.J. Sung, *Combust. Flame* 150 (2007) 355–368.

[8] M. Shukla, A. Susa, A. Miyoshi, M. Koshi, *J. Phys. Chem. A* 112 (2008) 2362–2369.
[9] V. Detilleux, J. Vandooren, *J. Phys. Chem. A* 113 (2009) 10913–10922.
[10] T.C. Zhang, L.D. Zhang, X. Hong, et al., *Combust. Flame* 156 (2009) 2071–2083.
[11] H.P.S. Shen, J. Vanderover, M.A. Oehlschlaeger, *Proc. Combust. Inst.* 32 (2009) 165–172.
[12] G. da Silva, C.C. Chen, J.W. Bozzelli, *J. Phys. Chem. A* 111 (2007) 8663–8676.
[13] S.J. Klippenstein, L.B. Harding, Y. Georgievskii, *Proc. Combust. Inst.* 31 (2007) 221–229.
[14] K. Hemelsoet, V. Van Speybroeck, M. Waroquier, *Chemphyschem* 9 (2008) 2349–2358.
[15] G. da Silva, J.W. Bozzelli, *J. Phys. Chem. A* 113 (2009) 12045–12048.
[16] G. da Silva, J.W. Bozzelli, *J. Phys. Chem. A* 113 (2009) 8971–8978.
[17] C. Cavallotti, M. Derudi, R. Rota, *Proc. Combust. Inst.* 32 (2009) 115–121.
[18] G. da Silva, J.W. Bozzelli, *Proc. Combust. Inst.* 32 (2009) 287–294.
[19] Y.Y. Li, F. Qi, *Acc. Chem. Res.* 43 (2010) 68–78.
[20] Y.Y. Li, L.D. Zhang, T. Yuan, et al., *Combust. Flame* 157 (2010) 143–154.
[21] Z.Y. Zhou, M.F. Xie, Z.D. Wang, F. Qi, *Rapid Commun. Mass Spectrom.* 23 (2009) 3994–4002.
[22] J.H. Kent, *Combust. Flame* 14 (1970) 279–281.
[23] R.A. Shandross, J.P. Longwell, J.B. Howard, *Combust. Flame* 85 (1991) 282–284.
[24] R.M. Fristrom, *Flame Structure and Processes*, Oxford, New York, 1995.
[25] A.T. Hartlieb, B. Atakan, K. Kohse-Hoinghaus, *Combust. Flame* 121 (2000) 610–624.
[26] R.J. Kee, J.F. Grcar, M.D. Smooke, J.A. Miller, Sandia Report SAND85-8240, Sandia National Laboratories, 1985.
[27] R.J. Kee, F.M. Rupley, J.A. Miller, Sandia Report SAND89-80098, Sandia National Laboratories, 1989.
[28] J.F. Grcar, R.J. Kee, M.D. Smooke, J.A. Miller, *Proc. Combust. Inst.* 21 (1988) 1773–1782.
[29] L.D. Zhang, J.H. Cai, T.C. Zhang, F. Qi, *Combust. Flame* 157 (2010) 1686–1697.
[30] H. Wang, X. You, A.V. Joshi, et al., 2007. Available from: <http://ignis.usc.edu/USC_Mech_II.htm>.
[31] H. Wang, M. Frenklach, *Combust. Flame* 110 (1997) 173–221.
[32] J. Appel, H. Bockhorn, M. Frenklach, *Combust. Flame* 121 (2000) 122–136.
[33] H. Richter, S. Granata, W.H. Green, J.B. Howard, *Proc. Combust. Inst.* 30 (2005) 1397–1405.
[34] R. Bounaceur, I. Da Costa, R. Fournet, F. Billaud, F. Battin-Leclerc, *Int. J. Chem. Kinet.* 37 (2005) 25–49.
[35] J.C.G. Andrae, T. Brinck, G.T. Kalghatgi, *Combust. Flame* 155 (2008) 696–712.
[36] G. Blanquart, P. Pepiot-Desjardins, H. Pitsch, *Combust. Flame* 156 (2009) 588–607.
[37] T. Seta, M. Nakajima, A. Miyoshi, *J. Phys. Chem. A* 110 (2006) 5081–5090.
[38] S.G. Davis, C.K. Law, *Combust. Sci. Technol.* 140 (1998) 427–449.
[39] J.P. Botha, D.B. Spalding, *Proc. R. Soc. Lond. A* 225 (1954) 71–96.
[40] Y.-R. Luo, *Comprehensive Handbook of Chemical Bond Energies*, CRC Press, Boca Raton, FL, 2007.
[41] G. da Silva, J.A. Cole, J.W. Bozzelli, *J. Phys. Chem. A* 113 (2009) 6111–6120.
[42] M. Frenklach, H. Wang, *Proc. Combust. Inst.* 23 (1991) 1559–1566.
[43] M.B. Colket, R.J. Hall, M.D. Smooke, Mechanistic Models of Soot Formation, United Technologies Research Center Report No. UTRC94-28, 1994.
[44] M.B. Colket, D.J. Seery, *Proc. Combust. Inst.* 25 (1994) 883–892.
[45] C.F. Melius, M.E. Colvin, N.M. Marinov, W.J. Pitz, S.M. Senkan, *Proc. Combust. Inst.* 26 (1996) 685–692.